ShuleMap — Kenya Schools Resource Mapping System

Planning, Analysis and Design Document

Version: 1.0 (MVP prototype)
Date: 25 September 2026
Status: Working prototype implemented and deployed (demo data)

1. Background and Problem Statement

Kenya has over 30,000 public primary schools and thousands of secondary institutions spread across 47 counties. Resource allocation decisions — electricity connections, boreholes, ICT labs, textbook supply, teacher deployment — are made at national and county level, but planners routinely lack a current, spatially organised picture of *which schools have what*. Data that exists sits in disconnected EMIS extracts, county spreadsheets and NGO project reports, and ages quickly because there is no lightweight channel for schools themselves to report changes.

The result: interventions are targeted from stale or aggregate data, gaps are discovered late, and schools that have already received support remain on priority lists while newly deprived schools go unseen.

ShuleMap addresses this with a live, map-first registry of school resources, an openly computable *Resource Adequacy Score*, and a moderated community reporting channel that keeps the map fresh between formal EMIS cycles.

2. Objectives

1. Provide a single public view of school resourcing (electricity, water, internet, ICT, library, science lab, textbooks, staffing) across all 47 counties.
2. Surface gaps objectively through a transparent, weighted Resource Adequacy Score, so counties can rank and prioritise.
3. Keep data fresh through submissions from teachers, head teachers, county officers and NGO partners, verified by county data officers before the map changes.
4. Expose data through an API so other systems (EMIS, county dashboards, NGO platforms) can consume it.

3. Stakeholders

Stakeholder	Interest	Role in system
Ministry of Education (MoE)	National planning, equity audits	Owner of master registry; consumes aggregated dashboards

Stakeholder	Interest	Role in system
County Education Offices	County-level allocation and verification	Moderators of community submissions
Head teachers / teachers	Accurate representation of their school	Submit resource-change reports
TSC (Teachers Service Commission)	Staffing deployment	Consumes staffing dimensions
NGO / donor partners (e.g. rural electrification, ICT programmes)	Targeting beneficiaries	Consume gap maps; submit completion reports
Parents and communities	Transparency	View school profiles
Kenyan Open Data initiative	Public data availability	Consumes open API

4. Analysis — User Stories (MVP scope)

- As a **county officer**, I can see all schools in my county on a map, colour-coded by need band, so I can identify the most under-resourced schools at a glance.
- As a **planner (MoE)**, I can see national KPIs (coverage of electricity, water, internet, ICT labs) and a ranking of counties by mean resource score, so I can allocate budgets where gaps are largest.
- As a **parent or community member**, I can search any school and see its full resource profile and how it compares.
- As a **teacher or head teacher**, I can report a change at my school (e.g. new borehole, solar installation, donated computers) in under two minutes, without an account.
- As a **county data officer**, I can review each submitted report, see the old → new diff, and approve (which updates the map and marks the school verified) or reject it with a note.
- As a **developer**, I can query the registry and scores through a JSON API.

Out of MVP scope: user accounts and authentication, bulk EMIS import, offline/ USSD submission, historical trend tracking, alerts. These are roadmap items (§10).

5. Functional Requirements

ID	Requirement	Priority
FR1	Registry of schools with geo-coordinates, level, gender, ownership, enrolment, staffing	Must
FR2	Resource inventory per school: electricity source, water source, internet, computers/lab, library, science lab, textbooks per pupil, SNE unit, feeding programme	Must
FR3	Computed Resource Adequacy Score (0–100) with per-dimension breakdown	Must
FR4	Interactive map of all schools, markers coloured by need band, drill-down to school profile	Must
FR5	Filters: county, level, ownership, minimum score, presence of power/water/internet/lab	Must
FR6	Searchable, sortable directory table of schools	Must
FR7	National dashboard: KPIs, county gap ranking, coverage radar, need-band distribution, priority list	Must

ID	Requirement	Priority
FR8	Anonymous submission of resource-change reports with proposed new values	Must
FR9	Moderation queue with approve/reject; approval applies changes to the school record and stamps verification date	Must
FR10	Read-only JSON API for schools, stats and submissions	Should
FR11	Export of filtered data (CSV)	Could (roadmap)
FR12	Accounts, roles and audit trail	Should (roadmap)

6. Non-Functional Requirements

- **Performance:** interactive map and dashboard usable on low-end Android devices and 3G connections; initial payload kept lean (single JS bundle, vector tiles).
- **Accessibility:** WCAG 2.1 AA — keyboard navigation, labelled form fields, colour never the sole carrier of meaning (markers paired with scores and text bands).
- **Data quality:** every change passes human moderation; each record carries a `lastVerified` date; demo/official data is clearly labelled.
- **Privacy:** no personal data of learners; submitter contact details are optional and used only for follow-up.
- **Availability:** static frontend on CDN; stateless API backed by SQLite (prototype) — production path to managed Postgres.

7. Data Model

Two entities in the MVP:

School — one row per institution:
`id, code (MoE-style code), name, county, subCounty, level (Primary/Secondary), gender (Mixed/Boys/Girls), ownership (Public/Private), enrollment, teachers, lat, lng, electricity (none/solar/generator/grid), water (none/river/rain/borehole/piped), internet (none/mobile/broadband), computers, computerLab, library, scienceLab, textbooksPerPupil, specialNeedsUnit, feedingProgram, lastVerified`

Submission — one row per community report:
`id, schoolId (FK), submitterName, submitterRole, contact (optional), message, resourceUpdates (JSON of proposed field changes), status (pending/approved/rejected), submittedAt, reviewedAt, reviewNote`

Resource Adequacy Score

Weighted checklist, 0–100 (higher = better resourced). Weights: electricity 18, water 14, internet 16, ICT 14, textbooks 14, staffing 14, library 10, science lab 10 (secondary only; weights renormalised for primary schools). Each dimension scores 0–1 against its target (e.g. textbooks per pupil ≥ 1.0, pupil:teacher ≤ 40:1 primary / 35:1 secondary, solar scores 0.7 of grid's 1.0). Bands: <30 critical, 30–49 high need, 50–69 moderate, ≥70 well resourced.

The formula is deliberately transparent and adjustable — counties can re-weight dimensions to match policy priorities without changing the data.

8. Architecture and Design

As built (prototype)

```
React 18 + Vite + Tailwind/shadcn SPA
  ├─ Dashboard  (recharts: bar/radar/donut + KPI cards)
  ├─ Map Explorer (MapLibre GL + OpenFreeMap tiles, custom DOM markers)
  ├─ Schools directory (sortable table)
  ├─ Report a Change (react-hook-form style flow)
  └─ Review Queue (moderation, diff view old → new)
        │  fetch (TanStack Query, hash routing)
        ▼
Express REST API  (/api/schools, /api/submissions, /api/submissions/:id/review)
  └─ Drizzle ORM → SQLite (data.db)
        ▲
Seed generator (deterministic, 235 demo schools across 47 counties)
```

Key design decisions:

- **MapLibre GL + OpenFreeMap** — no API keys, no billing, open data; performs well on low-end devices.
- **Crowdsourced updates are proposals, not writes.** Submissions only mutate the registry after a county officer approves; approval also stamps `lastVerified` — a simple but effective data-integrity workflow that production hardens with accounts and audit logs.
- **Score computed, not stored** — derived from raw dimensions at read time, so re-weighting is instantly retroactive.
- **Deterministic seed data** — clearly labelled synthetic demo data across all 47 counties (5 schools each, urban/arid resourcing gradients). Production replaces this with an official MoE/EMIS registry extract.

Production architecture (target)

SPA on CDN → API tier (Node/Express or serverless) → PostgreSQL with PostGIS for spatial queries; auth via Kenyan government identity (e-Gov) or phone OTP for submitters; county-level RBAC for moderation; scheduled EMIS reconciliation jobs; CSV/GeoJSON export endpoints; optional offline-first PWA and USSD short-code for low-connectivity submission.

9. Implementation Summary

The MVP was implemented end-to-end in this repository:

- `shared/schema.ts` — Drizzle schema for schools and submissions with zod insert validation.
- `shared/scoring.ts` — single source of truth for the Resource Adequacy Score (shared by API and UI).
- `server/seed.ts` — deterministic generator: 235 schools across 47 counties with urban/arid resourcing gradients, 3 seeded demo submissions.
- `server/routes.ts` — REST API with zod validation; approve applies whitelisted field updates to the school and stamps verification.
- `client/` — five pages behind a sidebar shell, dark mode, mobile layout, WCAG-minded markup.
- **QA** — 15-step Playwright end-to-end suite (all passing): marker rendering, filters, detail sheets, search/sort, full report → review → approve/reject cycle including verification that approved changes actually update the school record, dark mode, mobile overflow checks, zero console errors.

10. Roadmap

Phase	Scope
1 (done)	Registry, map, scoring, dashboards, moderated reporting — current prototype
2	Accounts (phone OTP), county RBAC, audit trail, CSV/GeoJSON export, full EMIS import
3	Historical snapshots and trend charts (score over time per school/county)
4	Intervention tracking: link donor projects to schools, measure gap closure
5	Offline PWA + USSD submission; open data portal integration (data.go.ke)

11. Risks and Mitigations

Risk	Mitigation
False or malicious reports	Moderation workflow (built), roadmap adds reputation scores and cross-checks against county records
Stale data between cycles	Community reporting channel; `lastVerified` surfaced on every profile
Data sovereignty concerns	Host in-country in production; align with Kenya Data Protection Act 2019; no learner-level personal data
Score gaming / misinterpretation	Transparent formula, per-dimension breakdown shown on every profile, weights adjustable per policy
Duplicate school records	MoE institution code as unique key; fuzzy-match on import (phase 2)

12. Demo Notes

All school names, coordinates and resource values in the deployed prototype are **synthetic demo data.** The map is centred on real county seats with jitter; OpenFreeMap tiles are real open data. To go live, replace the seed with an official registry extract — the schema and API need no changes.